

The image shows a person's hands typing on a laptop keyboard. Overlaid on the scene are several semi-transparent digital graphics. On the left, a flowchart with icons for a server, a bidirectional arrow, a cloud, and a gear is connected by lines. In the center, a large bar chart with a blue line graph is visible. Below it, a smaller bar chart and a pie chart are shown. At the bottom, a candlestick chart is partially visible. The background is a dark, blue-toned image of a person working on a laptop, with faint binary code (0s and 1s) scattered throughout.

E-commerce Sales Performance Analysis

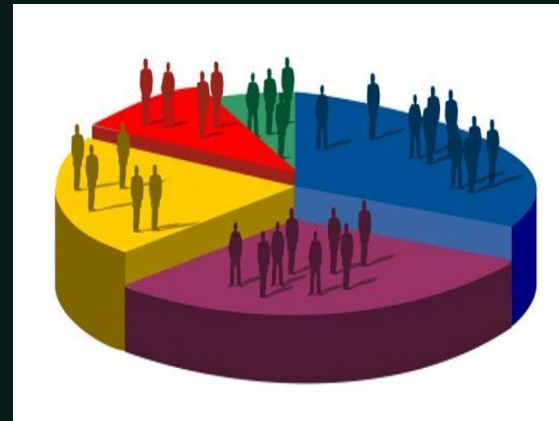
IVAN ZHUK

2025

Project Goals & Data Source

1. Analyze sales trends and product performance.
2. Assess logistics efficiency and customer satisfaction.
3. Identify key customer and seller demographics.
4. Apply ML techniques to forecast future sales or customer behavior.

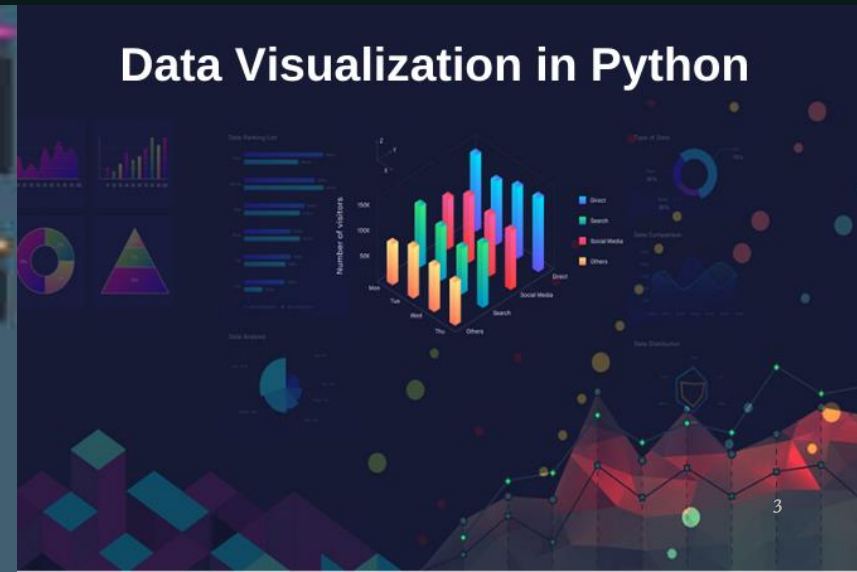
Source: Large-scale E-commerce dataset (Kaggle).



Methodology & Tools

Tools: Python (Jupyter Notebook, Pandas, NumPy), SQL (SQLite for data merging), **Visualization** (Matplotlib, Seaborn, Folium, Tableau). **Process:** Data Cleaning → Exploratory Analysis (EDA) → Time & Logistics Analysis → Rating & Demographic Analysis → Future Prediction.

```
1 # checking response.status_code (if you get 502, try rerunning the code)
2 if response.status_code != 200:
3     print(f"Status: {response.status_code} - Try rerunning the code")
4 else:
5     print(f"Status: {response.status_code}\n")
6
7 # using BeautifulSoup to parse the response object
8 soup = BeautifulSoup(response.content, "html.parser")
9
10 # finding Post images in the soup
11 images = soup.find_all("img", attrs={"alt": "Post image"})
12
13 # downloading images
```



Spellcheck &
Standardize
formats



Delete
duplicates



Exclude
irrelevant
observations



Handle
missing data



Address
outliers



Data Preparation & Cleaning

All 7 datasets were loaded, inspected, and cleaned. Key steps included: **Handling Missing Values** (especially in delivery dates and product categories). **SQL Joins** were used to create a unified analytical table for cross-dataset queries.

Business Context



Olist is the largest marketplace in Brazil (similar to Amazon Marketplace).



Dataset: 100K+ orders, 90K customers, 3K sellers, 2016–2018.

“How can data help improve sales, customer retention, and delivery performance?”



Median Price
Concentration

Total Orders (count)

Gross Freight Revenue (\$)

Gross Revenue (\$)

Average Check Price (\$)

Price sectors

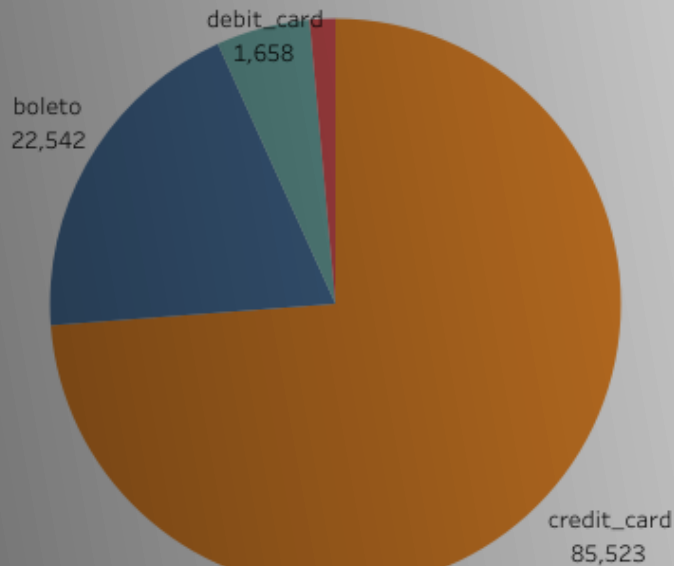
115,878

2,326,351

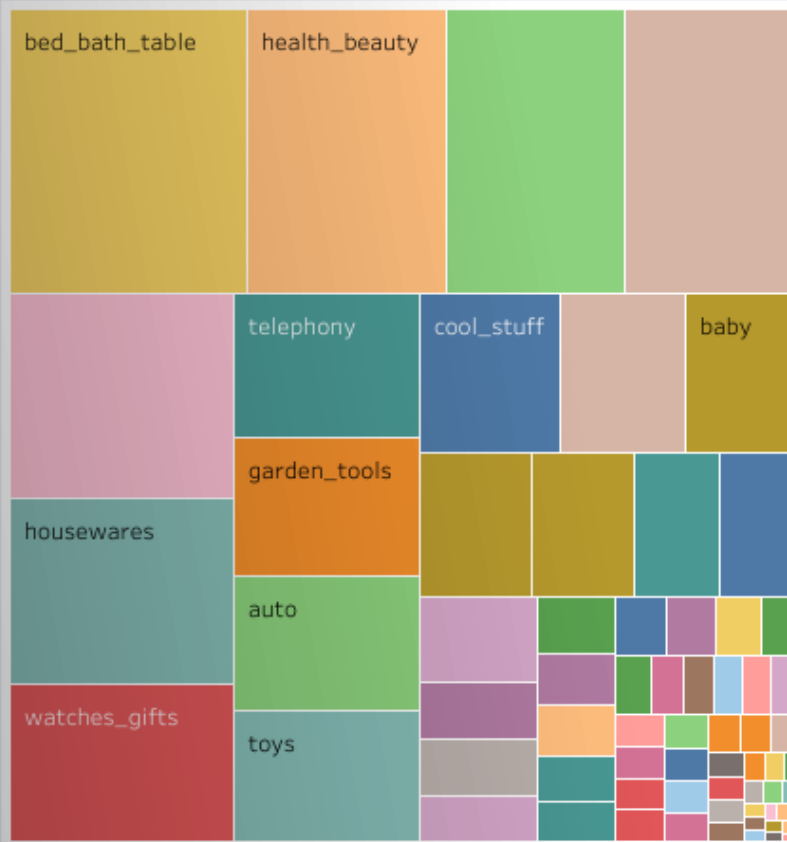
14,012,514

120.9

Payment Type



Product orders sorted by count of orders



Product category sorted by Total Revenue



Average check sorted by payment type



price

7000K
6500K
6000K
5500K
5000K
4500K
4000K
3500K
3000K
2500K
2000K
1500K
1000K
500K

Exploratory Analysis

MAIN FINANCIAL INDICATORS

Key Findings — Sales & Products

	Values
Total Orders	115,878
Gross Revenue	14,012,514
Average Check	120,9
Freight Revenue	2,326,351

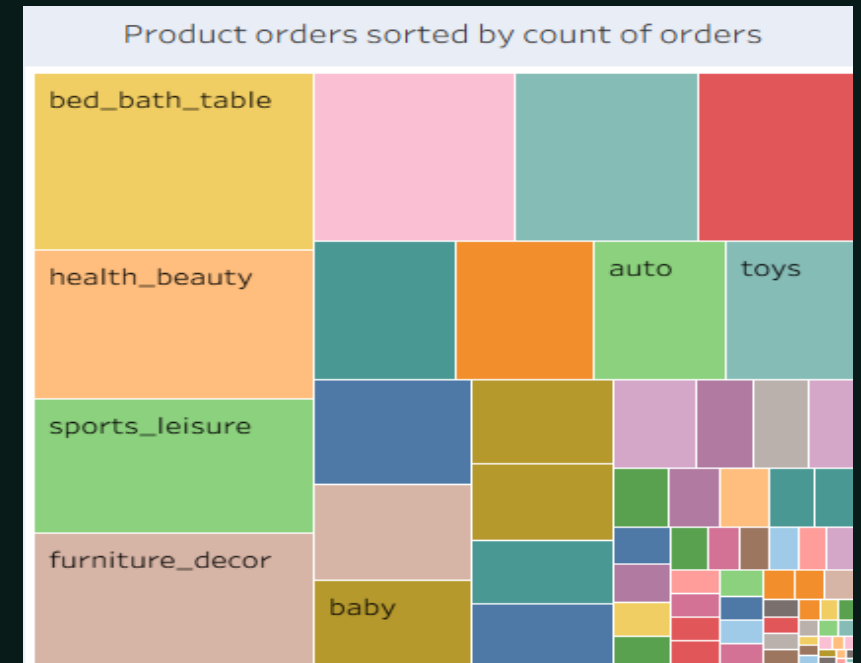
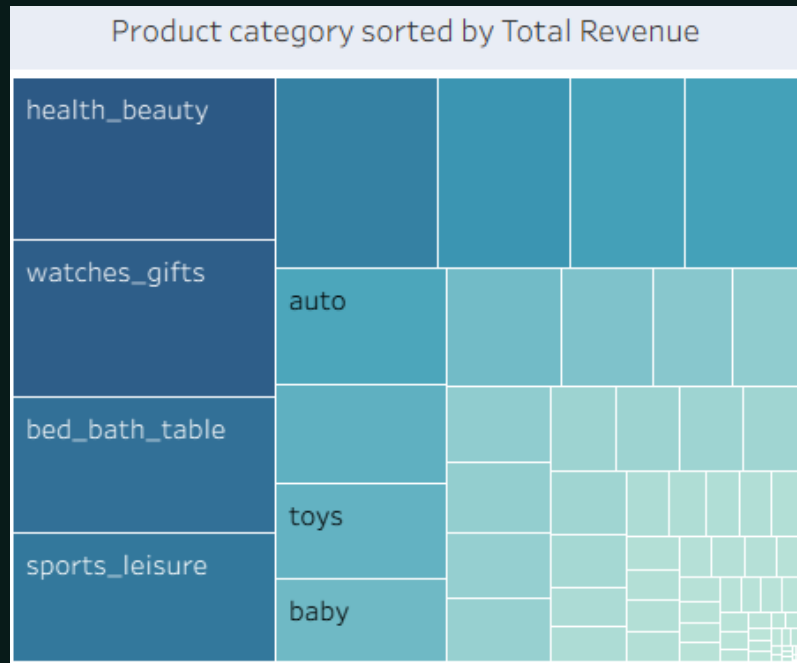
These indicators make it possible to look at the volume of information and provide
an initial basis for further research

Product Categories



Top Revenue: Health & Beauty, Watches
& Gifts, Auto, Sports & Leisure

Highest Volume: Bed & Bath, Health &
Beauty , Sports Leisure, Furniture &
Decor

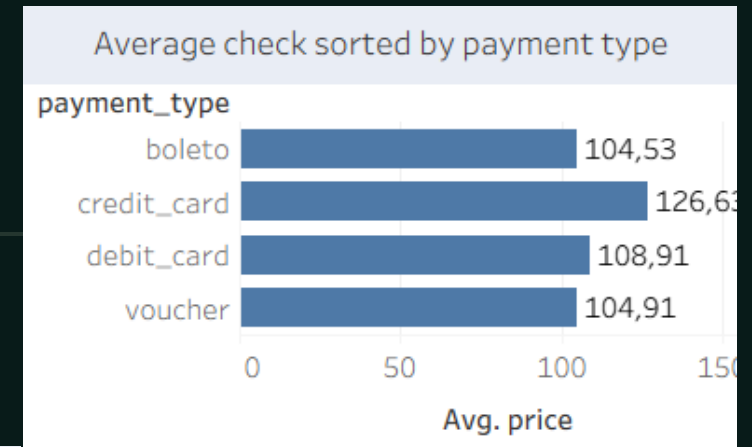


INSIGHT:

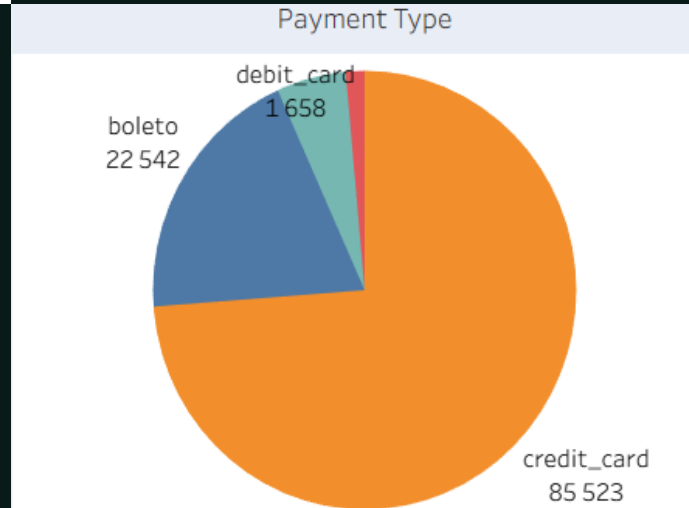
HIGH-VOLUME DOESN'T ALWAYS EQUAL HIGH-PROFIT — PREMIUM
CATEGORIES DRIVE REVENUE EFFICIENCY.



Payment Analysis



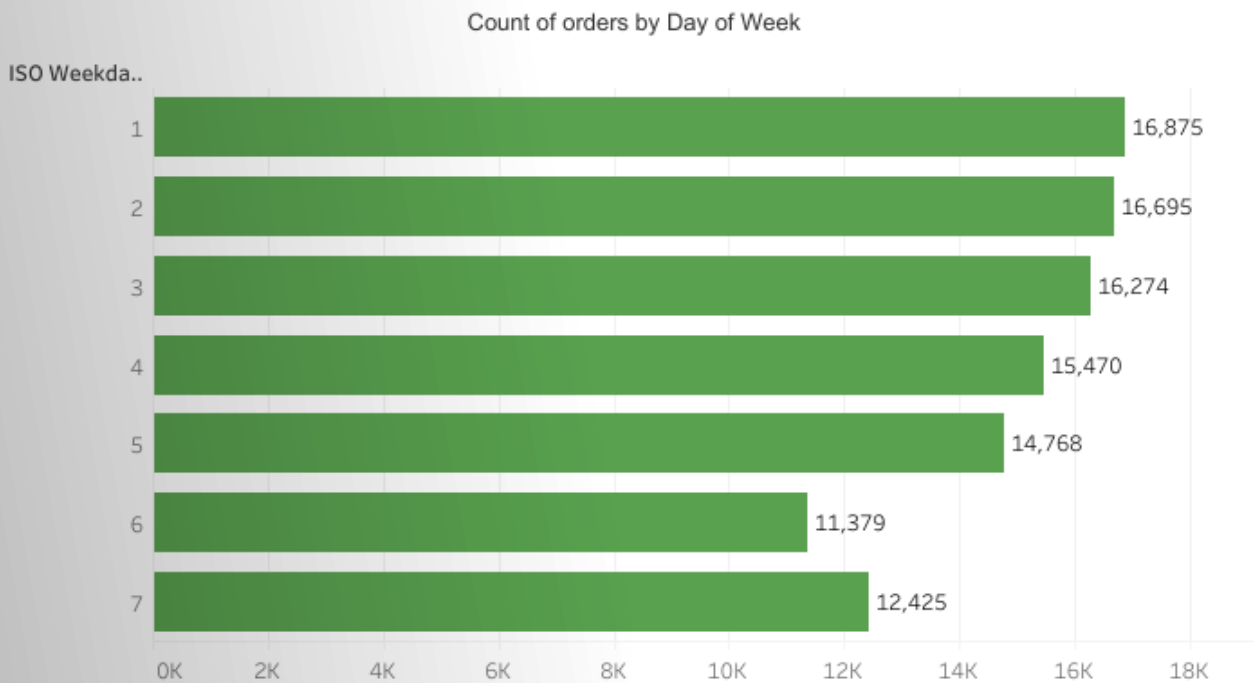
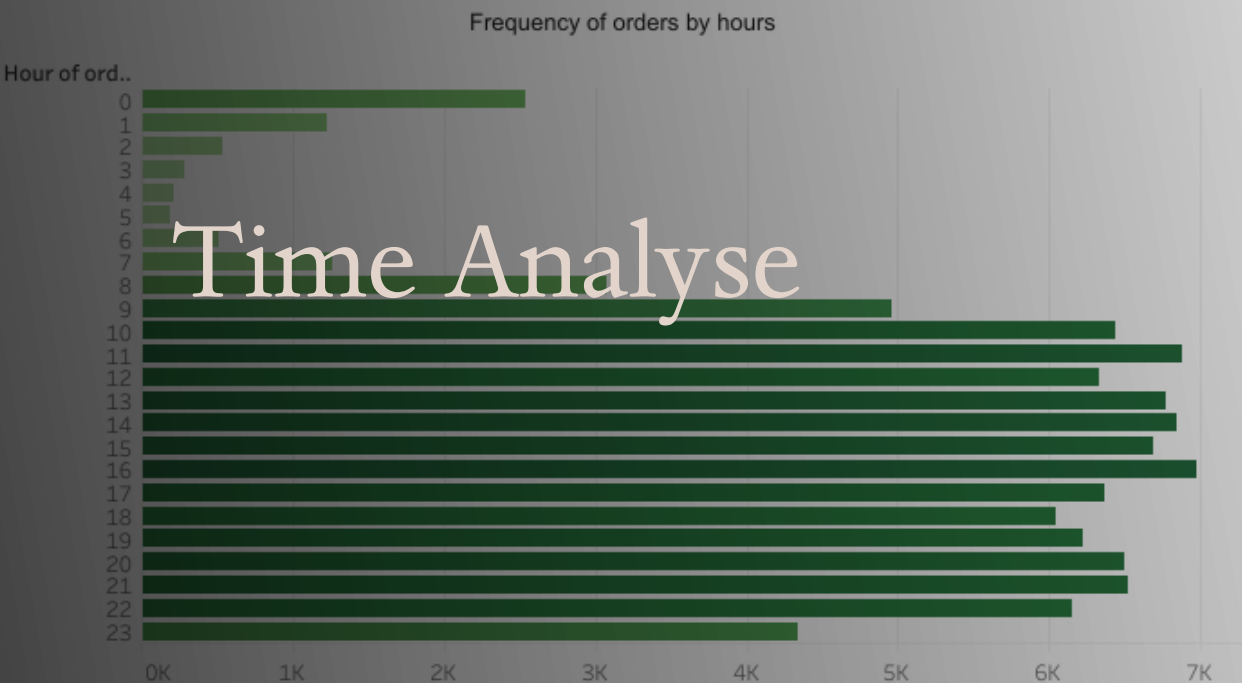
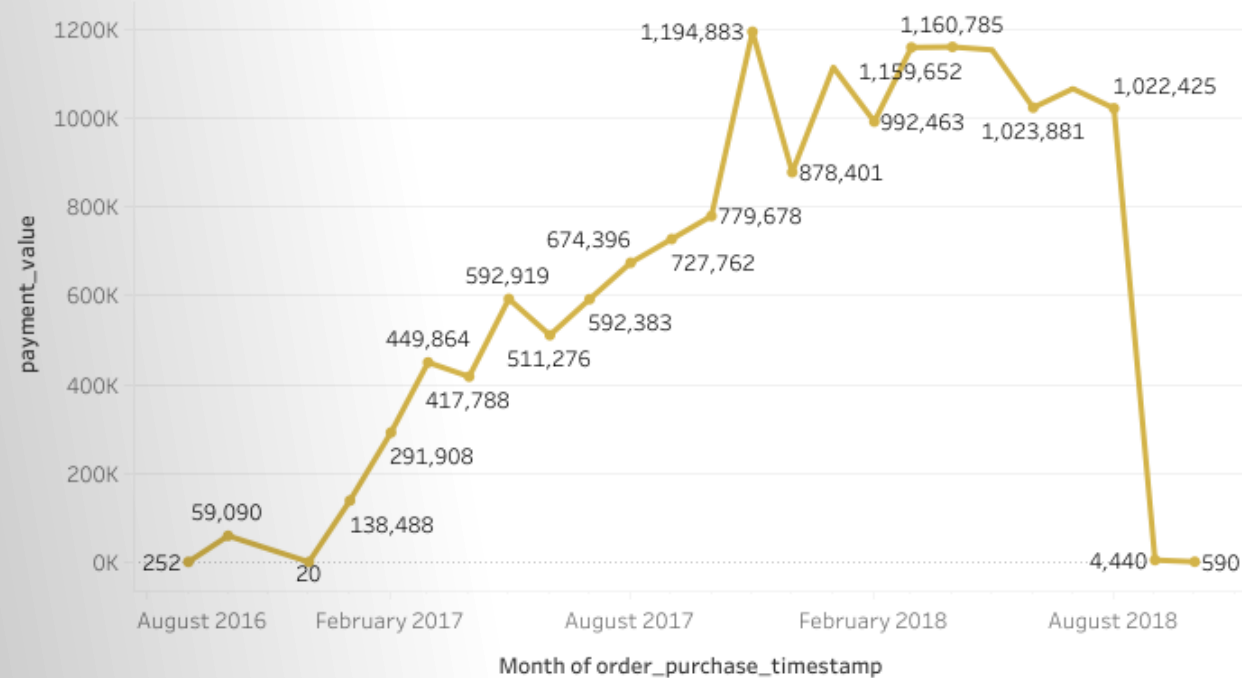
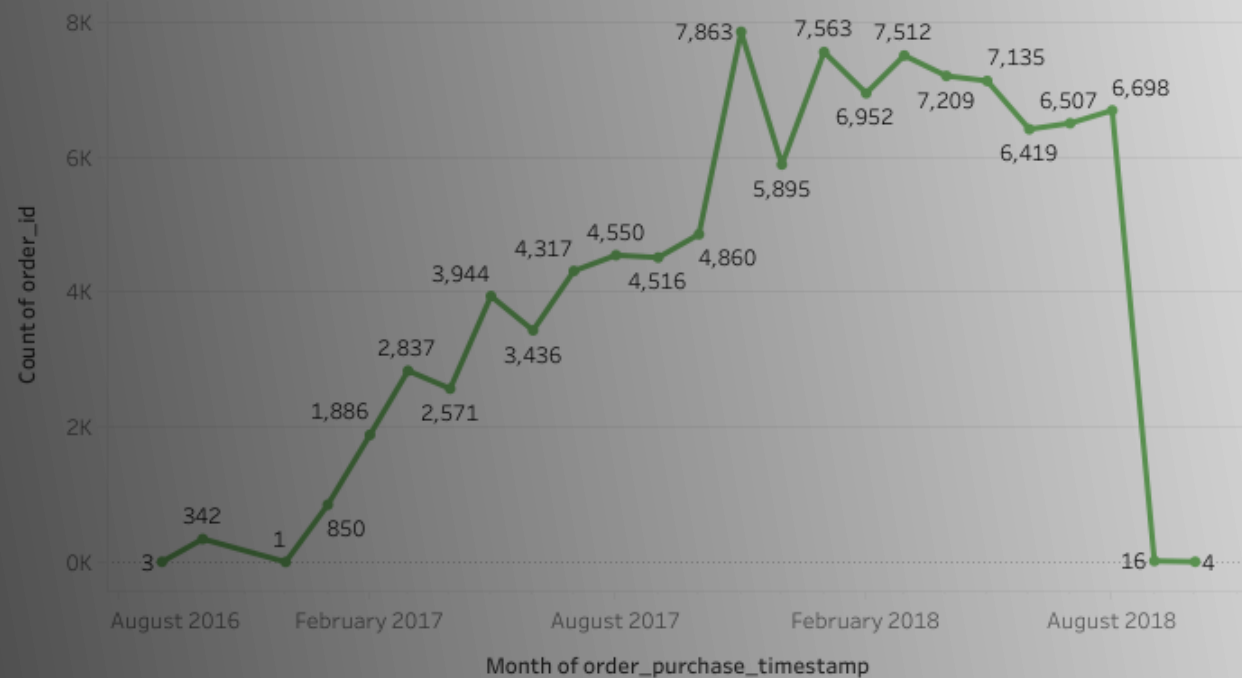
Credit Card: 74 % of transactions
 Boleto: 19 % of transactions
 Other: 7 % of transactions



Average check by method

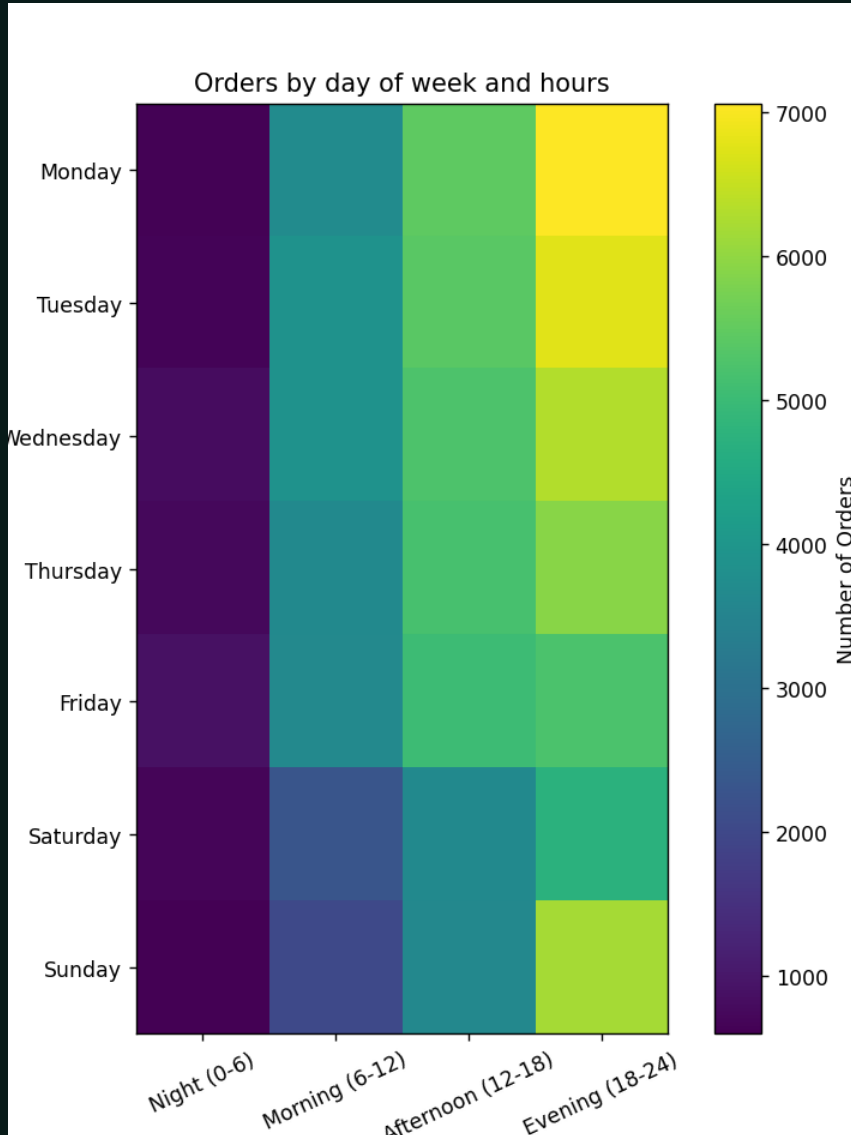
Credit Card → 126.6
 Voucher → 104,9
 Debit → 108.9
 Boleto → 104.5

CUSTOMERS USING CREDIT CARDS SPEND MORE — INCENTIVES FOR INSTALLMENTS
 COULD INCREASE AOV.



Time Analyse

Time Analyse



Peak time:

Peak days:

Seasonal peak:

Evenings (18:00–23:00)

Monday–Wednesday

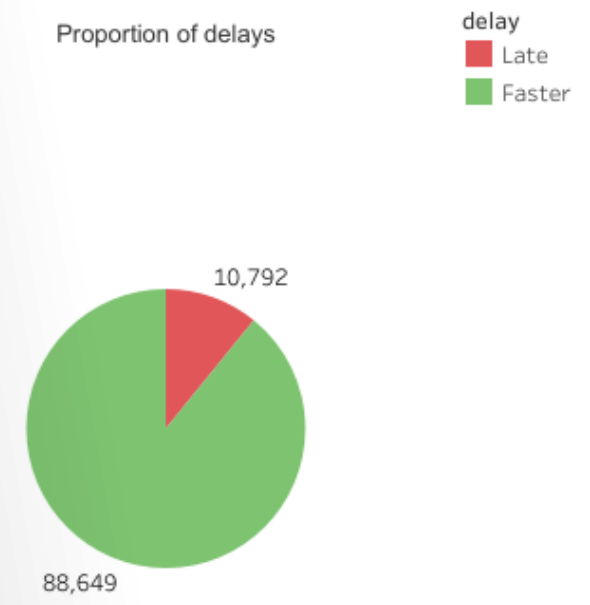
November (Black Friday)



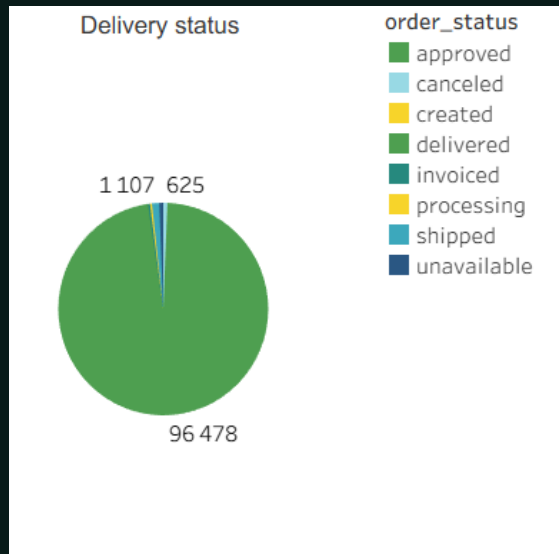
MARKETING CAMPAIGNS SHOULD FOCUS
ON EARLY-WEEK EVENINGS TO MAXIMIZE
CONVERSIONS.

■ real

Expected and actual delivery time



Delivery

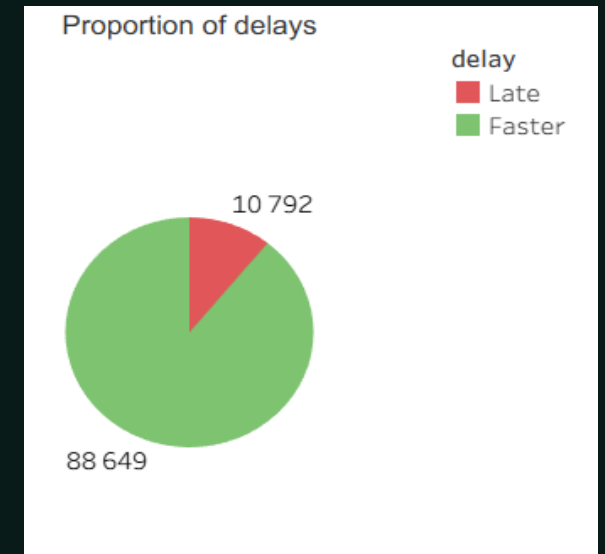


Avg delivery time: 12 days

Early deliveries: 25%

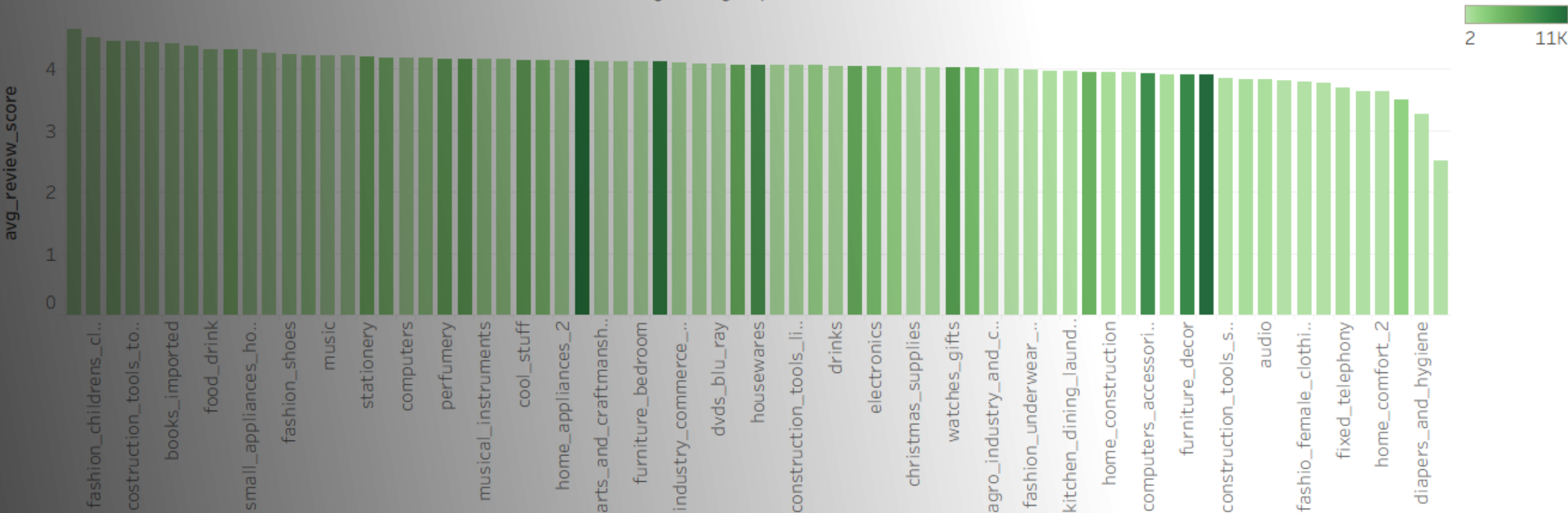
Late deliveries: 15%

Avg improvement vs estimate: 12 days earlier



Average Delivery Time: ~12 days. **Delivery Ahead of Schedule:** Orders are, on average, delivered **12 days faster** than the estimated date. **Delay Rate:** Only **6.73%** of orders were delivered late, showing high logistical reliability. **Finding:** There is a notable correlation (-0.26) between delivery delay and customer review score.

EXCELLENT LOGISTICS PERFORMANCE;
DELAYS STRONGLY INFLUENCE CUSTOMER
RATINGS.



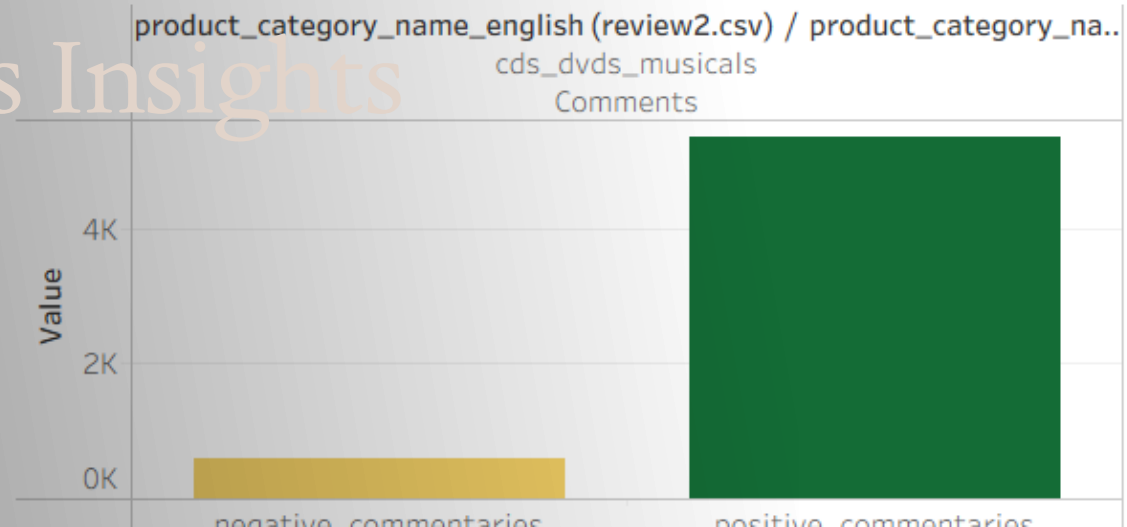
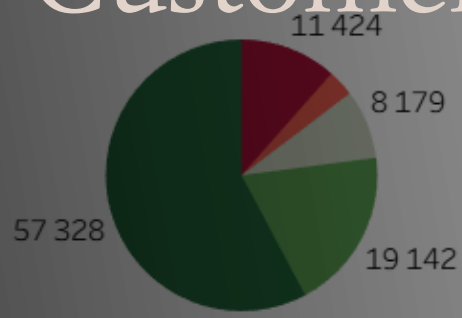
Distribution of ratings from customers

Count Reviews
99 224

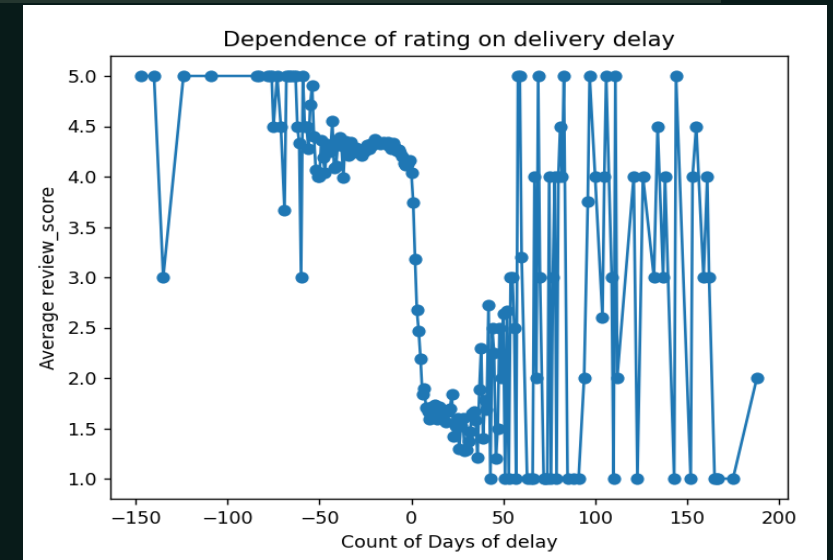
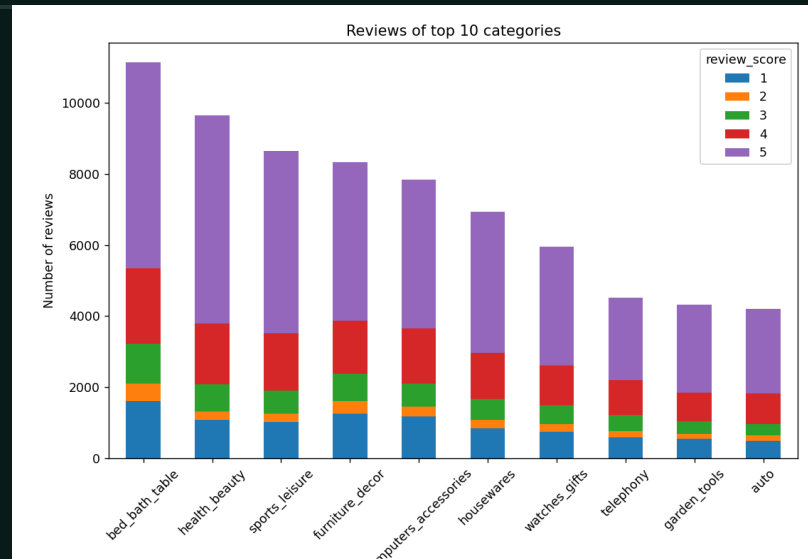
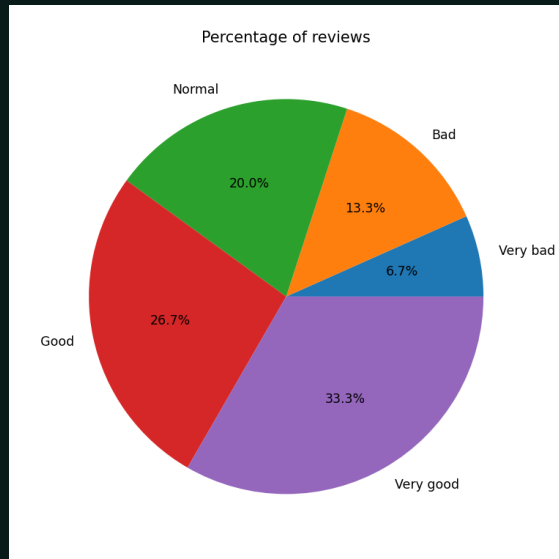
Positive and negative commentaries (from pattern words)

Measure Names
negative_comm..
positive_comme..

Customer & Ratings Insights



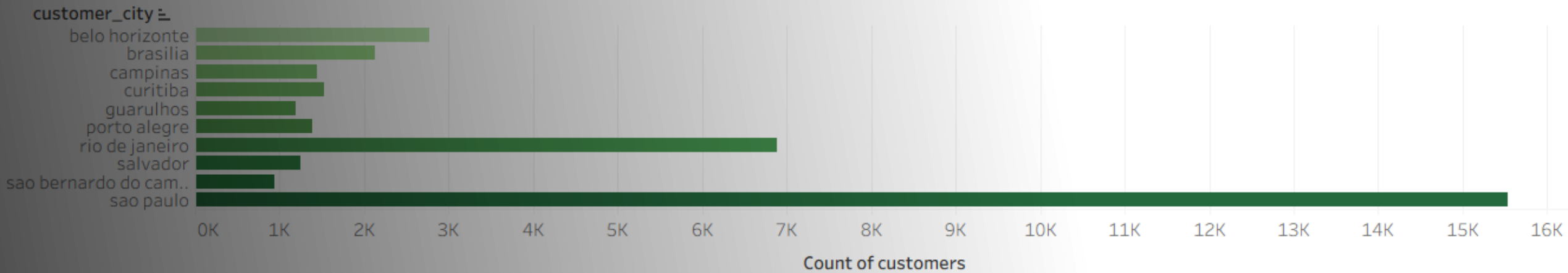
Review Analysis



Overall Satisfaction is High: 60% of reviews are positive (4 or 5 stars). **Polarization:** Reviews are highly polarized (mostly 1 or 5, very few 2 or 3). **Problem Areas:** Categories like **Security & Services** and **Fashion Male Clothing** have the lowest average ratings.

RATINGS ARE HEAVILY TIED TO DELIVERY
AND PACKAGING QUALITY. FAST DELIVERY
= HIGHER RATING.

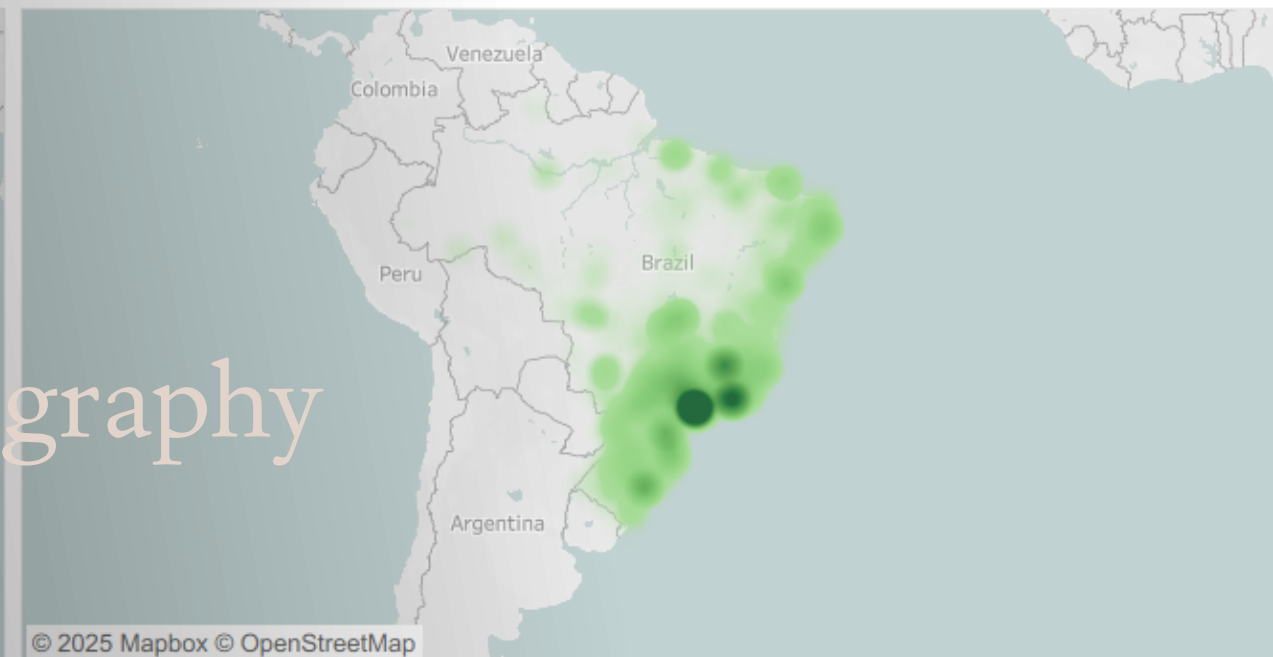
Top 10 City by customers



Concentration of sellers

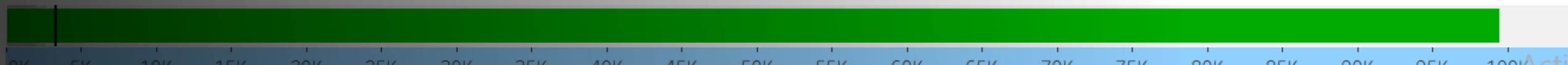


Concentration of orders

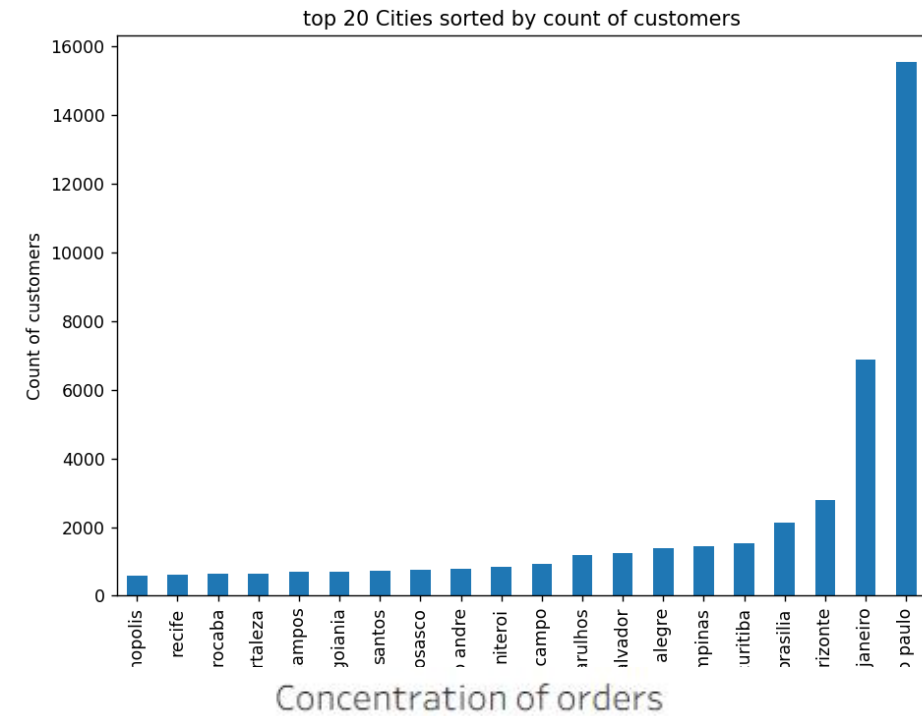


Customer & Seller Geography

Count of repeat Customers



Customer Geography



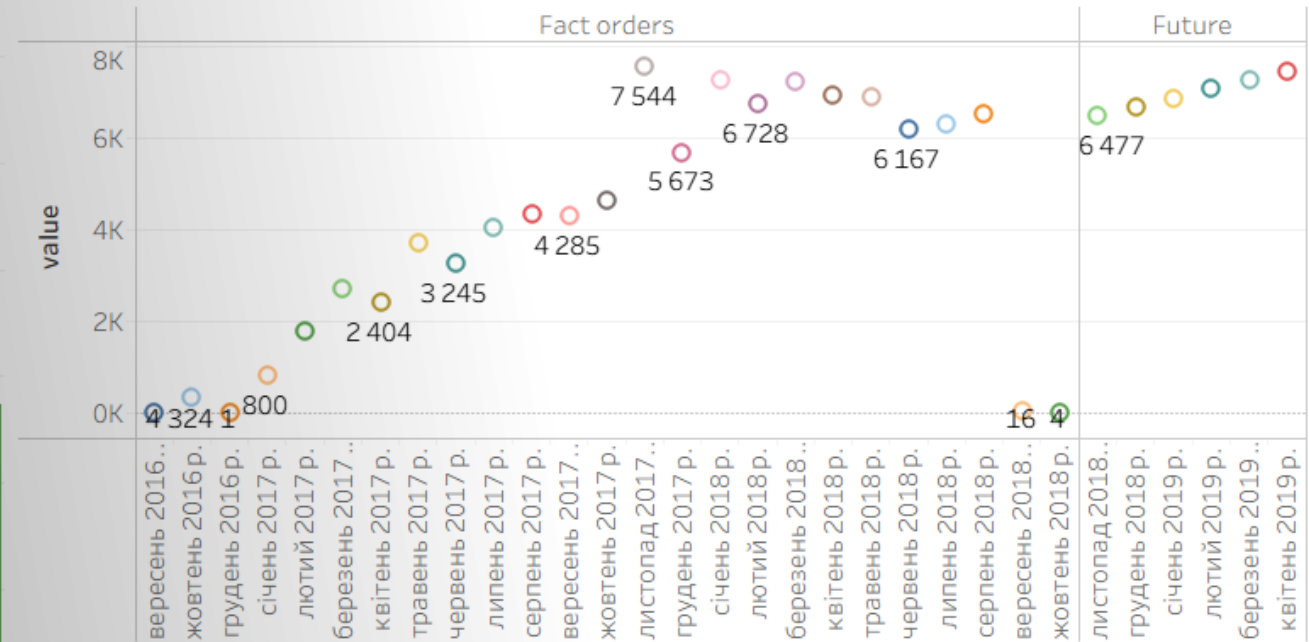
Customer Hubs: The majority of customers are concentrated in **São Paulo (SP)** and **Rio de Janeiro (RJ)**. **Seller Concentration:** Most sellers are also located in **SP** and **RJ**. **Insight:** This dictates optimal warehouse and logistics placement to serve the largest markets. **Competitive Market:** The Top 10 sellers account for only **12.3%** of the total sales revenue, indicating strong market competition and the absence of monopolists. **Low Repeat Rate:** Only **2.52%** of customers placed more than one order.

CUSTOMER BASE CONCENTRATED IN
SOUTHEAST — IDEAL AREA FOR
LOGISTICS OPTIMIZATION.

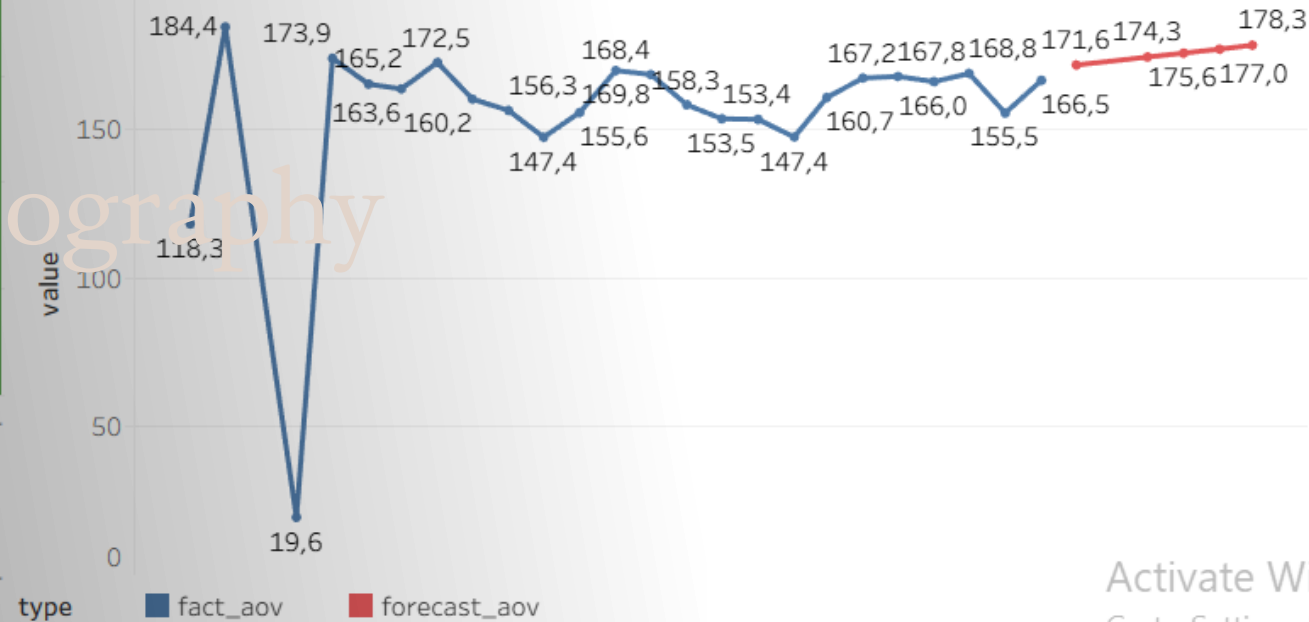
Average check prediction (mean)

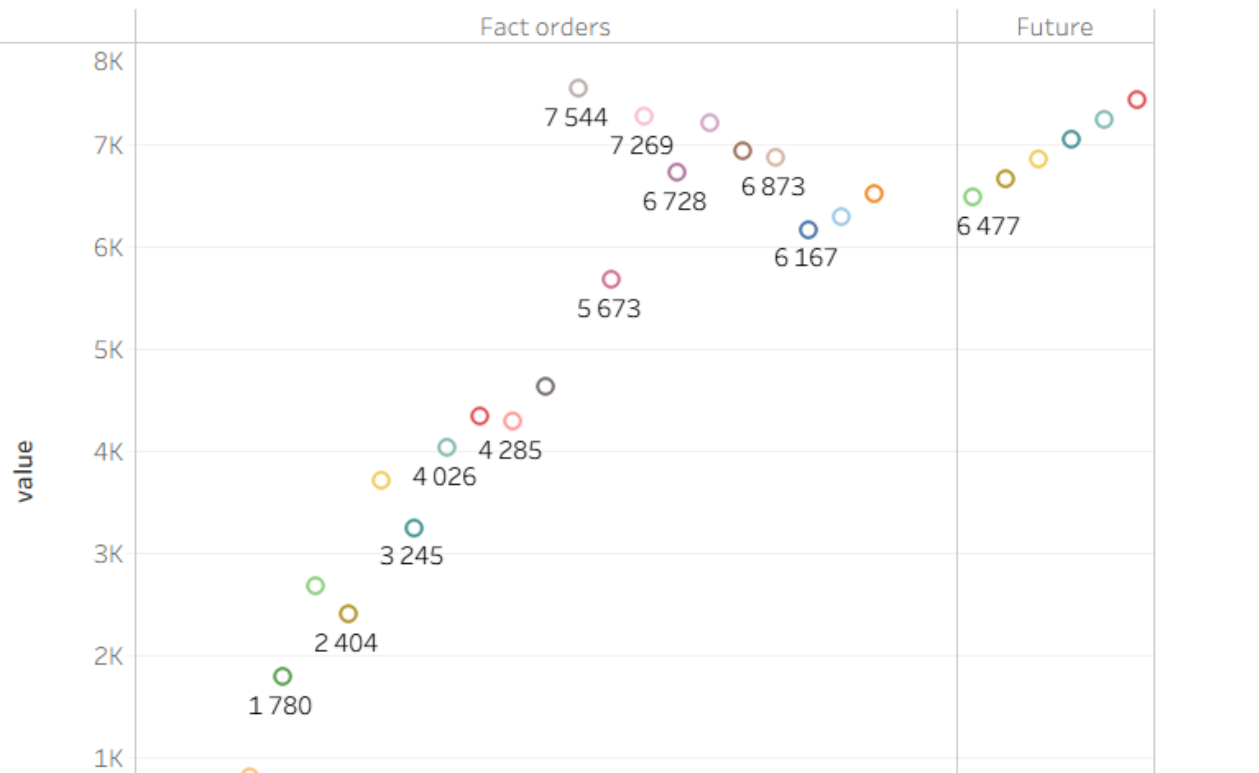
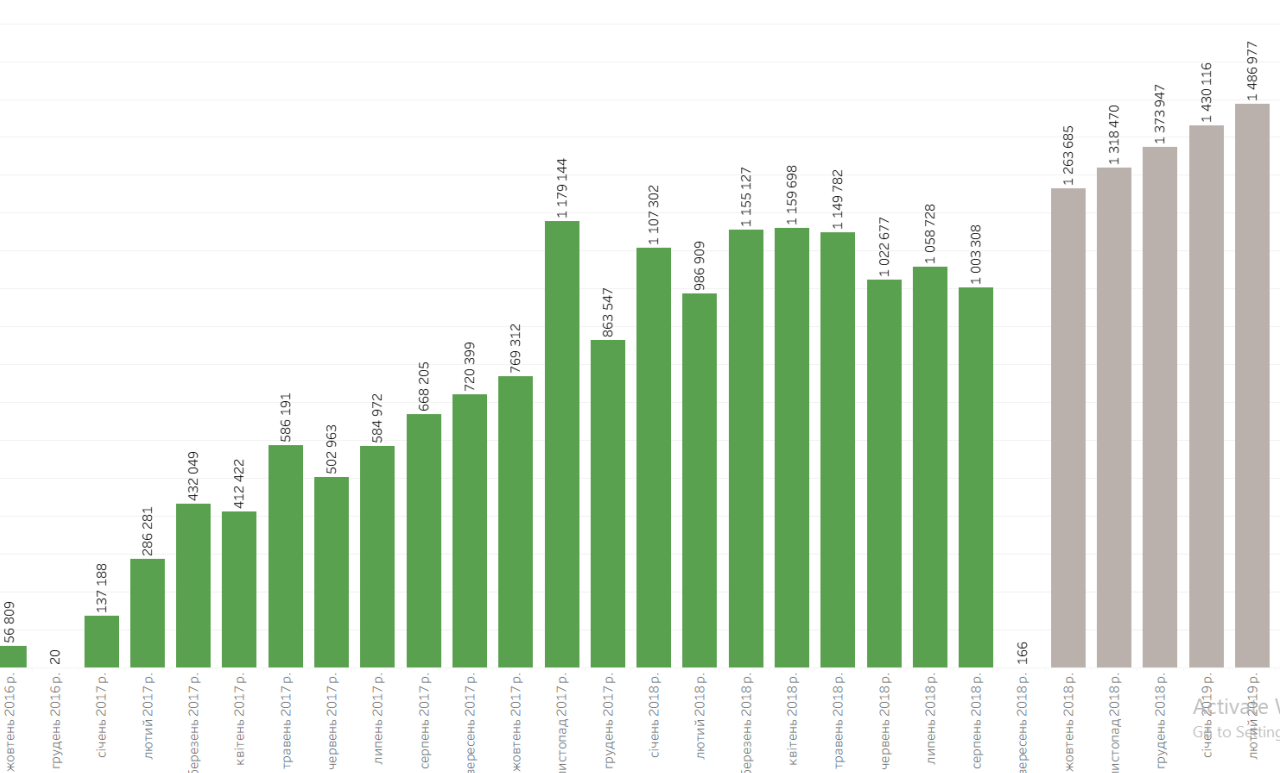


Orders fact and prediction for 6 Month



Average check prediction (Linear Regression)





- Linear Regression model
- $R^2 = 0.50$
- MAE = 174,609
- RMSE = 287,731
- Forecast horizon: 6 months
- Model shows steady upward trend, but variance remains high.

Forecasting Revenue

Methodology: Sales volume (Orders) and Gross Revenue were projected for the next six months using *Linear Regression and mean predict*. **Key Result — Revenue:** Gross Revenue is projected to demonstrate a consistent upward trend, reaching its peak in Month 6 at 1,544,530. **Key Result — Volume:** The Total Number of Orders is also forecasted to increase, reflecting continuous market expansion and customer retention. **Growth Rate:** The projected growth rates indicate **accelerated market penetration**, validating current strategic directions.

THE MODEL CAPTURES THE GENERAL GROWTH TREND BUT SHOULD INCORPORATE SEASONALITY FOR PRECISION.

Business Impact & Recommendations

1. **Leverage Logistics (Strength):** Maintain the significant competitive advantage of fast delivery, which currently surpasses estimated times by an average of 12 days.
2. **Address Loyalty (Opportunity):** Strategic focus is mandatory to increase the 2.52% repeat customer rate. This is the main growth bottleneck.
3. **Validate Forecasts:** Use the 6-month revenue and volume growth predictions to immediately optimize inventory and marketing budgets for peak demand (especially during evening hours and in the SP/RJ regions).
4. **Quality Assurance:** Investigate and resolve quality issues in low-rated categories (e.g., Security & Services) to support overall customer satisfaction.

Thank you

Q&A and Contact Information

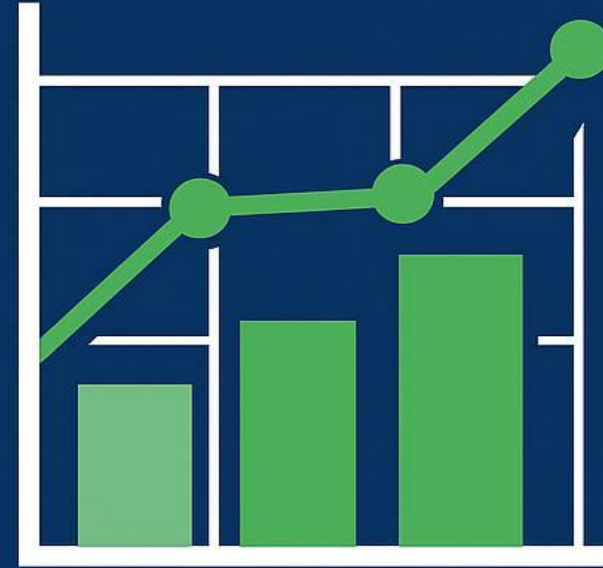
Thank you for your time. I am now open to your questions regarding the findings and the strategic recommendations.

Contact: Ivan Zhuk |

i.zhuk@gmx.de |

[https://github.com/Zhuk-](https://github.com/Zhuk-Ivan-S)

Ivan-S



IVAN ZHUK